Hippolyte Wallaert

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Education Background

Stanford University 2025 – 2027
M.S. in Computational and Mathematical Engineering (ICME) | GPA: 4.07/4.3 *Stanford, CA*
- Key courses: *Deep Learning Alchemy*, *Parallel Computing*, *Aligning Superintelligence*, *Explainable NeuroAI*, *Reinforcement Learning*, *Numerical Linear Algebra*, *ML with Graphs*.

École Polytechnique 2022 – 2025
Diplôme d'ingénieur with major in Applied Mathematics | GPA: 3.94/4.0 | Top 5 % *Palaiseau, France*
- Key courses : *Optimization & Control*, *Statistics*, *Machine/Deep Learning*, *Advanced Deep Learning*, *Operations research*, *Computer Vision*, *Stochastic calculus*, *Markov Chains & Martingales*, *Advanced Real Analysis*.

Research Experience

Mechanistic Interpretability 04/2026 – Present
Stanford University | Research Assistant | Advisor: Prof. Éric Darve *Stanford, CA*
- Investigating nonlinear representations and activation-steering methods for controlling large language model behavior.

Hard-Constrained Neural Networks 01/2026 – 03/2026
Stanford University | Research Project | Advisor: Prof. Madeleine Udell *Stanford, CA*
- Diagnosed gradient filtering in differentiable repair layers and developed an augmented-Lagrangian training approach to mitigate vanishing or exploding gradients under convex and nonlinear constraints.
- Presented a [poster] at the ICME Research Symposium.

Organ Allocation 12/2025 – 06/2026
Stanford University | Research Assistant | Advisor: Prof. Itai Ashlagi *Stanford, CA*
- Analyzed kidney transplantation data to study how allocation priorities and patient preferences shape organ-allocation outcomes.

Optimization & Operations Research 04/2025 – 08/2025
Massachusetts Institute of Technology | Visiting Student Researcher | Advisor: Prof. Alexandre Jacquillat *Cambridge, MA*
- Developed an exact algorithm for solving two-stage linear optimization problems with unknown cost and access to a zero-order oracle, building Benders-type cuts and exploiting algebraic dependencies to reduce oracle calls.
- Validated the method on facility-location instances motivated by robotic warehousing. Received highest honors from École Polytechnique's research internship committee. [Report]

Teaching experience

Teaching Assistant - Numerical Linear Algebra 09/2026 – Present
Stanford University, CME 302 *Stanford, CA*
- Refine assignments and exams and hold weekly office hours for a graduate course enrolling 100 students.

Physics Oral Examiner 09/2023 – 05/2024
Lycée privé Sainte-Geneviève *Versailles, France*
- Conducted weekly oral examinations for second-year preparatory students preparing for national entrance exams.

Industry Experience

Linagora 06/2024 – 08/2024
Software Engineer Intern *Grande Baie, Mauritius*
- Engineered a decentralized open-source Matrix.org messaging server using TypeScript.

Skills

Languages: English (fluent), French (native), Chinese (basic)

Programming: Python (PyTorch), CUDA, C++, JavaScript/TypeScript, Julia, Gurobi

Leadership and Activities

French Military Golf Team | National Gold Medalist & CISM World Team Gold Medalist 2024 – 2026

XForum (Career Fair) | Chief Organizer of Universities Fair (1,000 students and 40 universities). 2024

French Air Force 09/2022 – 03/2023
Assistant to Company Commander (Military Service) *Saint-Dizier, France*
- Trained 30 recruits for initial air commando training and led surveillance missions at Saint-Dizier Air Force Base.